

BINTULU, Malaysia

WMO: 964410

Lat: 3.124N

Lon: 113.020E

Elev: 23

StdP: 101.05

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	22.8	21.0	15.7	25.4	21.6	16.3	24.9	5.8	28.2	5.3	28.5	1.8	100	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.5	33.8	27.1	33.1	27.0	32.7	26.8	28.2	32.1	27.8	31.6	27.5	31.2	3.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	22.9	31.0	26.8	22.5	30.6	26.2	21.7	29.7	91.3	32.3	89.2	31.6	87.4	31.6	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.6	4.2	21.5	35.3	0.5	1.2	21.1	36.1	20.8	36.8	20.6	37.4	20.2	38.3		
			20.8	28.8	0.5	0.8	20.5	29.4	20.2	29.9	20.0	30.4	19.6	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.8	26.9	27.5	27.9	28.2	27.9	27.8	27.8	27.6	27.4	27.3	27.1	(6)
(7)		DBStd	1.01	0.94	0.86	0.84	0.88	0.98	1.01	1.04	1.11	0.98	0.92	0.76	0.82	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6391	522	474	541	536	563	536	553	551	527	538	519	531	(10)
(11)		CDD18.3	3349	263	241	283	286	304	286	294	293	277	280	269	273	(11)
(12)		CDH23.3	31641	2234	2110	2687	2859	3155	2838	2894	2852	2623	2558	2436	2393	(12)
(13)		CDH26.7	10960	625	584	889	1042	1213	1066	1112	1082	948	872	788	739	(13)
(14)	Wind	WSAvg	1.8	1.8	1.9	1.8	1.7	1.7	1.7	1.7	1.8	1.8	1.7	1.8	(14)	
(15)	Precipitation	PrecAvg	3806	452	279	276	266	229	252	248	300	290	352	422	447	(15)
(16)		PrecMax	5386	1281	802	555	568	472	612	586	679	550	766	659	780	(16)
(17)		PrecMin	2724	79	76	34	62	59	0	70	55	101	138	118	231	(17)
(18)		PrecStd	531	244	168	141	112	97	114	113	151	112	121	116	120	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.1	33.0	33.6	34.1	34.1	34.1	34.2	33.8	33.1	32.8	32.3	(19)
(20)			MCWB	26.9	26.8	26.9	27.4	27.3	27.4	26.9	26.6	27.0	26.8	27.0	27.0	(20)
(21)		2%	DB	31.2	31.2	32.1	32.9	33.2	33.1	33.1	33.1	32.8	32.2	31.9	31.8	(21)
(22)			MCWB	26.6	26.5	26.6	27.2	27.1	27.1	26.7	26.8	26.8	26.5	26.6	26.7	(22)
(23)		5%	DB	30.4	30.6	31.2	32.0	32.5	32.2	32.2	32.2	32.0	31.7	31.1	31.0	(23)
(24)			MCWB	26.2	26.2	26.4	26.7	26.9	26.6	26.5	26.5	26.4	26.4	26.4	26.5	(24)
(25)		10%	DB	29.8	29.8	30.7	31.2	31.8	31.6	31.6	31.5	31.1	30.8	30.4	30.1	(25)
(26)			MCWB	25.8	25.9	26.3	26.5	26.7	26.5	26.4	26.3	26.3	26.2	26.1	26.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.6	28.2	28.5	28.4	28.6	28.4	28.5	28.3	27.7	27.9	27.8	(27)
(28)			MCD B	31.2	30.9	31.3	32.0	32.7	32.9	32.6	32.5	32.3	31.8	31.3	31.3	(28)
(29)		2%	WB	27.0	26.9	27.3	27.7	27.7	27.8	27.6	27.6	27.5	27.2	27.2	27.2	(29)
(30)			MCD B	30.3	30.2	30.7	31.5	31.7	31.7	31.6	31.5	31.3	30.9	30.5	30.6	(30)
(31)		5%	WB	26.5	26.5	26.8	27.2	27.4	27.2	27.1	27.1	27.0	26.7	26.7	26.7	(31)
(32)			MCD B	29.5	29.5	30.1	30.8	31.2	31.0	31.0	30.9	30.7	30.2	29.9	29.9	(32)
(33)		10%	WB	26.1	26.1	26.4	26.7	26.9	26.7	26.6	26.6	26.5	26.3	26.3	26.2	(33)
(34)			MCD B	28.8	28.9	29.5	30.1	30.6	30.3	30.3	30.2	30.0	29.7	29.4	29.2	(34)
(35)	Mean Daily Temperature Range	MDBR	6.4	6.3	7.0	7.3	7.5	7.7	7.8	7.9	7.5	7.3	7.1	6.8	(35)	
(36)		5% DB	MCD BR	7.4	7.3	7.8	8.3	8.6	8.8	8.7	8.9	8.8	8.4	7.9	7.7	(36)
(37)			MCWBR	3.3	3.3	3.4	3.5	3.6	3.8	3.7	3.9	4.0	3.7	3.5	3.5	(37)
(38)		5% WB	MCD BR	6.9	6.8	7.3	7.8	8.0	8.3	8.3	8.5	8.2	7.8	7.5	7.3	(38)
(39)			MCWBR	3.4	3.3	3.5	3.6	3.6	4.0	3.9	4.0	4.0	3.7	3.6	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.416	0.429	0.430	0.412	0.408	0.412	0.417	0.478	0.483	0.484	0.414	0.407	(40)	
(41)		taud	2.470	2.418	2.412	2.462	2.482	2.477	2.455	2.263	2.248	2.249	2.465	2.501	(41)	
(42)		Ebn at Noon	909	901	896	894	877	862	862	826	837	841	903	912	(42)	
(43)		Edh at Noon	114	123	123	115	109	108	111	138	143	143	114	109	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.53	4.94	5.34	5.39	5.09	4.93	5.04	5.10	4.92	4.72	4.71	4.59	(44)	
(45)		RadStd	0.35	0.30	0.23	0.21	0.20	0.23	0.22	0.20	0.32	0.34	0.20	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) +0.59	(h) +0.47	(i) +0.29	(j) N/A	(k) N/A	(l) +112	(m) +114	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.32	+0.26	+0.21	N/A	N/A	+66	+66	(47)

Nomenclature: See separate page